I tried to leverage Python networks package for executing these tasks. Overall, nodes represent authors and edges are connections of authors who coauthored a paper together.

For problem 1, the code helps calculate how degrees, the number of connections each author has, are distributed across the network. The degree distribution is plotted with a log-log graph as plot1.pdf. In hw1solution.txt file, the format @ 1 j nj, where j is the degree and nj is the number of nodes with that degree.

For problem 2, the code helps defines the largest connected component. It also is plotted with a log-log graph as plot2.pdf. The format @ 2 j kj in hw1solution.txt file represents that, j is the size of the component and kj is the number of components of that size. The ratio represents the size of the largest component compared to the total number of nodes.

For problem 3, the code results in the demonstration of nodes in the network from the root node, “Gries”. The format @ 3 j rj, where j is the distance from the root node, rj is the number of nodes at that distance.

For problem 4, the code transforms the average number of potential parents for nodes at each distance. The format @ 4 j pj, where j represents distances, and pj is the average number of potential parents.

There are some issues when handling with the data. First of all, some invaild values in the dataset need to be highlighted. Lines that are not representing vaild years will be skip to avoid errors. The chosen range, 1985 to 2005, needs to be filted, and this decision helps to hold a consistent analysis. Also, the log-log figures help plot reveal the trends because the original distribution of coauthorship networks usually showcase exponentially. Specifically in problem 3 and 4, if we miss Gries in the largest component, the first available node will be the root for the breadth-first search trees.

One paticular problem is that the nodes with no connections probably are not visualized in the distribution, which means the result will underestimate the number of authors in the network.

Reference:

https://networkx.org/documentation/stable/tutorial.html

https://networkx.org/documentation/stable/reference/functions.html

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